

Zhufeng (Zephyr) Qiu

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RESEARCH INTERESTS

GPU Systems | Compression-Communication Co-Design | High-Performance Computing | ML Systems | Distributed LLM Training & Inference

RESEARCH SUMMARY

I am interested in building efficient and scalable systems for large-scale ML/AI workloads, with a focus on GPU runtime optimization, collective communication, and distributed execution. I have worked on parallel data processing, production-scale distributed systems, and performance engineering across serial CPU, OpenMP, MPI, CUDA, and NCCL backends, including a lossless homomorphic compression for NCCL AllReduce payloads and a controlled study of when communication-computation overlap helps. My Ph.D. interests are in HPC and ML systems, particularly communication-efficient execution and high-performance LLM training and inference.

EDUCATION

Northeastern University, Khoury College of Computer Sciences

Jan. 2023 - May 2025

- M.S. in Computer Science, GPA: 4.0/4.0
- Relevant coursework: Parallel Data Processing, Algorithms, Computer Networking, Cloud Computing.
- Teaching Assistant for Algorithms and Parallel Data Processing.

University of Southern California, Viterbi School of Engineering

Aug. 2019 - May 2021

- M.S. in Applied Data Science, GPA: 3.83/4.0
- Relevant coursework: Scientific Computing and Visualization, Analysis of Algorithms, Machine Learning, Data Mining, Database Systems.

Wuhan University, School of Resource and Environmental Sciences

Sep. 2014 - Jun. 2018

- B.S. in Geographical Information Science, GPA: 3.58/4.0
- Relevant coursework: Data Structures, Linear Algebra, Discrete Mathematics, Probability and Mathematical Statistics, C Programming.

SELECTED SYSTEMS, HPC & ML PROJECTS

BoundRelay — LLM KV-Cache Compression and Transfer Study [GitHub](#)

Aug. 2026 - Sep. 2026

Independent ML-systems study | Python, PyTorch, CUDA, cuSZp, SZ3, CUDA zfp

- Replicated K/V compression-sensitivity asymmetry in Qwen3-1.7B on **32 held-out articles**: at similar payload sizes, K-only compression raised mean teacher-forced NLL by **0.0155 (95% CI [+0.0068, +0.0235])**, while V-only compression did not.
- Built a benchmark covering **1,152 tensor observations**, with recorded model/codec revisions and binary hashes, frozen evaluation inputs and tolerances, and a versioned corrections register.
- Ran **360 paired prepared-cache comparisons**: approximately 3× payload compression cut fsync-acknowledged write time by **26–49%**, but increased pipelined host-staged GPU-to-GPU transfer time to **1.52–2.33× raw**. Host staging bypassed peer-copy failures that returned CUDA success without transferring data on all three rented multi-GPU pods.
- Identified per-tensor overhead and payload-dependent bandwidth as sources of prediction error; stage costs measured on the actual cache and hardware predicted **21.43 ms versus 43.65 ms** for the complete pipeline.

Multi-GPU Similarity Engine - Lossless Homomorphic Collective Compression & Overlap [GitHub](#)

Jul. 2026 - Aug. 2026

Independent systems study extending USC INF 553 Data Mining / CSCI 596 Scientific Computing Projects | C++, CUDA, NCCL, MPI, OpenMP, PySpark, CMake, Nsight Systems

- Compressed the NCCL AllReduce payload 3× **losslessly (56.2 → 18.8 MB)** by packing six integer sufficient statistics into two uint64 words with carry-free fields, letting NCCL **sum them in compressed form without decompressing**; verified bit-exact across 2.58M pairs.
- Isolated the interconnect by running the identical binary on **NVLink and PCIe**: communication is 11% vs 91% of the iteration, so the same compression reduces the iteration time by **4% on NVLink and 55% on PCIe**, and combined with overlap cuts the PCIe iteration **2.53×**.
- Re-engineered a Yelp collaborative-filtering workload from Spark into serial C++, OpenMP, MPI, CUDA, and dual-GPU NCCL backends, keeping all five bit-exact; processed **1.17M candidate pairs in 4.21 ms on two NVIDIA A100s, 312× faster than serial and ~20× faster than 16-thread OpenMP**.
- Derived the overlap ceiling **min(compute, comm)/total** and verified it with Nsight traces; measured the largest overlap benefit when compute and communication times were comparable, and found and fixed a double-buffering race through a chunk-size sweep.

Quantized LLM Inference Study on a LoRA-Fine-Tuned Llama 3.1 8B [GitHub](#)

Jul. 2025 - Sep. 2025; Jul. 2026 - Aug. 2026

Independent inference-systems study extending an LLM-engineering course fine-tune | Python, PyTorch, CUDA, PEFT/LoRA, bitsandbytes, Hugging Face Transformers, Modal

- Identified a **2.50× end-to-end regression in the deployed configuration**: a bf16 compute dtype carried over from Ampere training onto Turing, which has no bf16 tensor cores. Localized entirely to prefill (4.42×, MFU collapsing to 3.2%) and isolated by an unquantized control agreeing within 0.8%.
- Showed **4-bit quantization buys capacity, not speed**: NF4 decode runs 1.69× slower than fp16 while moving 3.9× fewer bytes, reaching 9.8% of peak bandwidth against fp16's 64.1% (dequantization-bound, not bandwidth-bound).
- Built a CUDA-event inference benchmark across four quantization configurations on **Turing and Ampere**, decomposing latency into prefill and decode against an analytical roofline; showed **97% of the tokens are prefill but account for only 39% of total time**, with decode costing 54× more per token.
- Fine-tuned Meta Llama 3.1 8B with QLoRA (rank 32 / α 64 on attention projections, **27M trainable parameters, 0.34% of base**) under completion-only supervision on a curated 400,000-item dataset, and served it 4-bit NF4 on an NVIDIA T4 via Modal.

Zephyr Manus - Agent Execution and Tool-Orchestration Platform [Live Demo](#) **Mar. 2026 - May 2026**
Independent Project | FastAPI, Redis Streams, PostgreSQL, Docker, Next.js, MCP/A2A

- Built a Planner-ReAct agent runtime supporting task decomposition, external-tool execution, persistent state, and human-in-the-loop wait/resume workflows.
- Developed Docker-isolated execution for Browser, Shell, and File tools and an event-driven FastAPI backend using Redis Streams, PostgreSQL, SSE, and MCP/A2A.

Distributed Apriori Mining on Yelp Dataset [GitHub](#) **Feb. 2024 - May 2024**
Northeastern CS 6240 Parallel Data Processing | Java, Hadoop MapReduce, HBase, Hive, AWS S3

- Built a distributed Apriori pipeline on Hadoop MapReduce over ~550K Yelp reviews and ~15K businesses, decomposing candidate generation, support counting, pruning, and iterative frequent-itemset discovery, with Hive external tables preprocessing raw JSON into per-user review baskets.
- Implemented two interchangeable storage backends for cross-pass state: an HBase path materializing each pass as a table with support filtering applied at write time, and a file-based path over HDFS/S3 with a runtime flag switching between local and S3 discovery of per-pass outputs.

Train Ticket Flash-Sale System [GitHub](#) **Jan. 2025 - Sep. 2025**
Independent Project | Java, Spring Boot, Spring Cloud, Redis, RocketMQ, MySQL

- Built a high-concurrency ticket-purchasing service using Redis-based inventory control and RocketMQ-based asynchronous processing to decouple request handling from downstream order creation.
- Designed idempotency and consistency mechanisms for concurrent requests, mitigating duplicate purchases, overselling, and retry-related failures under flash-sale workloads.

RESEARCH EXPERIENCE

PM2.5 Inversion and Spatiotemporal Analysis for the Wuhan Metropolitan Area **Mar. 2016 - Oct. 2017**
National Undergraduate Innovation and Entrepreneurship Training Program | Core Team Member | Advisor: Prof. Huanfeng Shen

- Integrated ground-monitoring measurements, NASA MODIS/AOD satellite observations, and meteorological data, performing preprocessing and spatiotemporal matching across multiple data sources.
- Contributed to the development and evaluation of a mixed-effects model for PM2.5 inversion, using cross-validation and baseline comparisons to assess predictive performance.
- Used MATLAB/R for modeling and ArcGIS to generate spatially continuous PM2.5 concentration maps across the Wuhan “1+8” metropolitan area; the national program passed final review with a **Good** rating.

Web-Based Dynamic Thematic Mapping Technologies **2018**
Undergraduate Thesis | SinoMaps Press, Beijing | Advisors: Prof. Haihong Zhu and Yuntao Long

- Surveyed web-based thematic-mapping methods and designed the overall, functional, and logical architecture for online visualization, interaction, and dynamic data updates.
- Built and validated a D3.js customs trade-map and map-service prototype supporting thematic visualization, statistical charts, interactive queries, and live data updates.

INDUSTRY EXPERIENCE

Beaconfire Solutions Inc. - Software Development Engineer **Jul. 2025 - Present**
Seattle, WA (Remote)

- Developed enterprise HR and digital-commerce microservices with Spring Boot/Spring Cloud, integrating authenticated APIs, relational/document data stores, AWS S3, and RabbitMQ-backed asynchronous workflows.

NSFOCUS Technologies Group Co., Ltd. - R&D Engineer **Jun. 2021 - Dec. 2022**
Wuhan, China

- Built and maintained distributed security-data pipelines using Kafka and Elasticsearch for multi-source log ingestion and normalization, and implemented Spark/MapReduce batch workflows serving 100+ enterprise clients and processing approximately 5M records per day.
- Developed host-anomaly detection features for illicit file transfers and backdoor processes using fuzzy inference, hierarchical clustering, and k-core graph mining over multidimensional security logs.
- Contributed to a production alerting workflow that achieved effective response for **100% of severe threats and 90% of high-risk threats** in a telecom deployment.

NSFOCUS Technologies Group Co., Ltd. - R&D Engineer Intern **Jul. 2018 - Mar. 2019**
Wuhan, China

- Processed and normalized multi-source security logs covering user operations, file access, and process behavior to support downstream host-anomaly analysis.

TECHNICAL SKILLS

HPC / GPU: CUDA, NCCL, MPI, OpenMP, Nsight Systems, CMake
Languages: C/C++, Python, Java, Scala, SQL, JavaScript/TypeScript
ML / LLM: PyTorch, Hugging Face Transformers, PEFT/LoRA, bitsandbytes, scikit-learn, LangGraph, ReAct, RAG, MCP/A2A
Distributed / Data: Spark/PySpark, Hadoop MapReduce, Hive, HBase, Kafka, Elasticsearch, Redis, PostgreSQL, MySQL, Chroma, AWS
Systems / Software: Docker, Modal, Spring Boot, Spring Cloud, Git, Jenkins, Maven

HONORS

- Outstanding Student Leader, Wuhan University, 2014-2015 and 2015-2016.
- Outstanding Individual in Theoretical Study, Wuhan University, 2014-2015.